Kivotos Federal University · Fall 2026

CS 340: Software Design

Course Syllabus · 3 Credits

Instructor	Professor Plana
Email	plana@kivotos.edu (please include "CS 340" in the subject line)
Office Hours	Tuesdays 3:00–4:30 PM and Fridays 10:00–11:00 AM, Millennium Science Building 2.18
Meetings	Tuesdays & Thursdays, 1:30–2:45 PM, Millennium Science Building 1.04
Prerequisites	CS 240 (Data Structures) with a C or better
Platforms	MomoLearn for grades and quizzes; GitHub Classroom for all project submissions

Course Description

CS 340 teaches how to design software that is easy to understand, change, and test. Topics include requirements analysis, UML modeling, object-oriented design principles (SOLID), the classic design patterns, software architecture, refactoring, and design for testability. Projects are written in Java; the final project is completed in teams of three or four.

Learning Objectives

- Translate stakeholder needs into requirements, use cases, and UML class and sequence diagrams.
- Apply SOLID principles to evaluate and improve object-oriented designs.
- Select and implement appropriate creational, structural, and behavioral design patterns.
- Identify code smells and refactor safely with the support of automated tests.
- Communicate and defend design decisions in writing and in a team presentation.

Grading

Component	Weight
Quizzes (5, in class, 20 minutes each; lowest dropped)	15%
Project 1: Requirements Spec & UML Models (individual)	10%
Project 2: Design Patterns Implementation (individual)	12%
Project 3: Refactoring a Legacy Codebase (individual)	13%
Project 4: Team Application (design document 5%, design review 5%, final 15%)	25%
Final Exam (cumulative)	25%

Assessment Calendar

Quizzes are taken in class at the start of the period. Project submissions are due on GitHub Classroom by **11:59 PM** on the due date unless noted.

Due Date	Assessment	Type	Where / Time
Thu, Sep 17	Quiz 1: Requirements & UML	Quiz	In class, 1:30 PM
Fri, Sep 25	Project 1: Requirements Spec & UML Models	Project	GitHub Classroom, 11:59 PM
Thu, Oct 1	Quiz 2: SOLID Principles	Quiz	In class, 1:30 PM

Due Date	Assessment	Type	Where / Time
Fri, Oct 16	Project 2: Design Patterns Implementation	Project	GitHub Classroom, 11:59 PM
Thu, Oct 22	Quiz 3: Creational & Structural Patterns	Quiz	In class, 1:30 PM
Fri, Nov 6	Project 3: Refactoring a Legacy Codebase	Project	GitHub Classroom, 11:59 PM
Thu, Nov 12	Quiz 4: Behavioral Patterns & Architecture	Quiz	In class, 1:30 PM
Fri, Nov 13	Project 4 Milestone: Team Design Document	Project	GitHub Classroom, 11:59 PM
Tue, Nov 17	Project 4 Milestone: Design Review Presentations	Presentation	In class, 1:30 PM
Thu, Dec 3	Quiz 5: Testing & Design for Testability	Quiz	In class, 1:30 PM
Fri, Dec 4	Project 4 Final: Team Application & Demo Video	Project	GitHub Classroom, 11:59 PM
Tue, Dec 8	Final Exam (cumulative)	Exam	Millennium Science Building 1.04, 1:00–3:00 PM

Project Descriptions

Project 1: Requirements Spec & UML Models

Given an interview transcript with a fictional client, write a requirements specification with at least eight use cases, a domain model, and two sequence diagrams.

Project 2: Design Patterns Implementation

Extend a starter inventory system by applying at least three design patterns (one creational, one structural, one behavioral). Include a short design rationale explaining each choice.

Project 3: Refactoring a Legacy Codebase

Write characterization tests for a provided legacy application, then refactor it to remove at least six identified code smells without changing behavior. Submit a before/after design report.

Project 4: Team Application

Teams design and build a small application of their choice. The **Team Design Document** (architecture, UML, patterns used) is due first, followed by a 10-minute **Design Review Presentation** in class, and finally the working **application and a 5-minute demo video**.

Weekly Schedule

Week	Tuesday	Thursday
1	**Sep 8**: Course intro; what makes a design good?	**Sep 10**: Requirements, use cases
2	**Sep 15**: UML class and sequence diagrams	**Sep 17**: Domain modeling **Quiz 1: Requirements & UML**
3	**Sep 22**: Cohesion, coupling, encapsulation	**Sep 24**: SOLID: S, O, L
4	**Sep 29**: SOLID: I, D; design by contract	**Oct 1**: Creational patterns I **Quiz 2: SOLID Principles**
5	**Oct 6**: Creational patterns II	**Oct 8**: Structural patterns I

Week	Tuesday	Thursday
6	**Oct 13**: Structural patterns II	**Oct 15**: Behavioral patterns I
7	**Oct 20**: Behavioral patterns II	**Oct 22**: Layered and MVC architecture **Quiz 3: Creational & Structural Patterns**
8	**Oct 27**: Code smells	**Oct 29**: Refactoring catalog
9	**Nov 3**: Refactoring with tests	**Nov 5**: Event-driven architecture
10	**Nov 10**: Dependency injection	**Nov 12**: Team design workshop **Quiz 4: Behavioral Patterns & Architecture**
11	**Nov 17**: Design reviews **Project 4 Milestone: Design Review Presentations**	**Nov 19**: API design
12	**Nov 24**: Unit testing and test doubles	**Nov 26**: Thanksgiving Break, no class
13	**Dec 1**: Design for testability	**Dec 3**: Course review **Quiz 5: Testing & Design for Testability**
Finals	**Dec 8**: **Final Exam**, 1:00–3:00 PM	

No class Thursday, November 26 (Thanksgiving Break).

Course Policies

Late projects. Each student has three late days for the semester, usable on Projects 1–3 (one day = 24 hours). After late days are used, late work loses 10% per day. Late days cannot be used on Project 4 milestones.

Missed quizzes. The lowest quiz is dropped to cover an absence. Additional missed quizzes require documentation.

Collaboration. Discussing ideas is encouraged; sharing code on individual projects is not. AI coding assistants may be used for Project 4 only and must be documented in the README.

Team contribution. Project 4 grades are adjusted by a confidential peer evaluation submitted with the final project.